GS EXTERIOR EXPERTS · DECK DIVISION · PERMIT SET

5549 S Lansing Ct · Elevated Deck

5549 S Lansing Ct, Englewood · Rachel Chiu · 8' x 12' deck, 10' above the yard, off the existing door · TrueFrame 2x10 @ 16" on a (2)4x10 Douglas fir beam and (2) 6x6 DF #1 posts on Diamond Piers · TimberTech Reserve Driftwood, single picture frame, CONCEALoc · Impression Rail Express, vertical balusters, modern top rail, Bronze · October 17, 2026

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Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale. This set carries no pricing.

G-001 GENERAL NOTES · DESIGN CRITERIA

Loads, codes, structure summary, sheet index, site — the permit-set cover

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

GSX · Jade Pender 303-550-9558 · October 17, 2026 · Rev 0 · PERMIT SET

PROJECT

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech Reserve Driftwood, IRX bronze)
5549 S Lansing Ct, Englewood CO 80111
Prepared for Rachel Chiu
92 SF · 10'-0" above grade · 1 zone

DESIGN CRITERIA

Deck live load	40 psf uniform (IRC Table R301.5); guards 200 lb concentrated / 50 plf
Ground snow	30 psf (ground; deck design uses the larger of snow and live) — 40 psf live governs; 30 psf ground snow, roof 30 psf non-reducible (Arapahoe County design criteria); deck tables at 40 psf live + 10 psf dead
Dead load	10 psf deck
Roof snow (cover)	—
Design total	50 psf on the deck
Wind	115 mph Vult (3-second gust), Exposure B (suburban), Risk Cat. II — county design-criteria table (2015 cycle; confirm the 2021 table)
Frost depth	3'-0" (footings bear below)
Soil bearing	1,500 psf presumptive (IRC R401.4.1) — no soils report
Seismic	B
Wildfire (WUI)	not in a WUI zone
Codes	2021 IRC / IBC as adopted by Arapahoe County (Res. 21-394, eff. 4/1/2022); IRC R507 decks with the county-amended Colorado ICC Deck Building Guide; ASCE 7-16 loads; Tables R507.5 / R507.6 / R507.4 / R507.9
Engineering	Stamped structural set by a Colorado PE required — the stamped set governs

STRUCTURE

Footings	DP-50/50
Posts	2 x 6x6 Douglas fir #1 with 4x4 knee braces
Beams	DROP (2)4x10 DF#1: 7'-8", 2 posts at 9" / 7'-3" (worst 2,214 lb)
Joists	7 2x10 #1 True Frame Joist SYP GC @ 16" OC x 11'-1"
Rims	2-ply 2x10 front and sides
Ledger	2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners
Decking	TimberTech Reserve Driftwood — 24 rows parallel to the house, picture frame in Reserve Driftwood, 3/16" gaps, CONCEALoc
Rail	IRX Baluster 36" Bronze: 5 bays, 6 posts, 31 LF

SHEET INDEX

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D-501	Details

SITE

Rear / front lot line: deck edge 13.0' from the line, setback per the PUD...
Side lot line: deck edge 23.7' from the line, 5' required
Easement: Check plat R1535678 for rear / side utility and drainage...
Unincorporated Arapahoe County (Cherry Creek CDP; Englewood mailing address). Parcel 2075-14-4-05-001 / PIN 031772427, Lot 41 Blk 9 Cherry Creek Vista 6th Flg Amend Resub (plat R1535678), 6,401 SF corner lot (front S Lansing Ct east, street side E Berry Ave north). R-P PUD Cherry Creek Vista (Z75-030 / P75-027 / P75-028). Deck on the WEST (rear) wall, north end, hung on the 2' second-floor cantilever that runs the full 33' wall (assessor sketch). Tax: CO 2.9 + RTD 1.0 + CD 0.1 + Arapahoe 0.25 = 4.25% (no city).

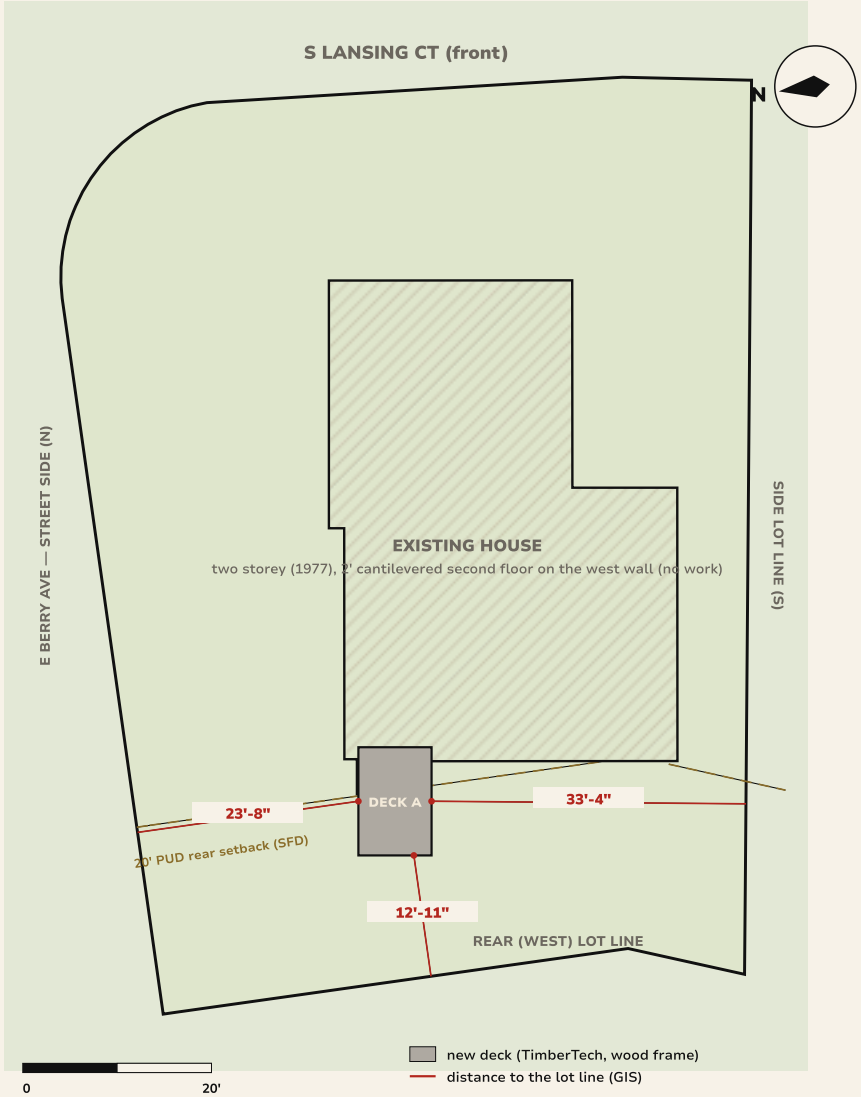
Elevated deck · C-100 · Site plan

C-100 **SITE PLAN**

Parcel, existing house, new decks and roof covers, distances to the lot lines, coverage

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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PARCEL

5549 S Lansing Ct, Englewood CO 80111
Parcel 2075-14-4-05-001 · PIN 031772427
Lot 41, Block 9, Cherry Creek Vista 6th Filing Amended Resub (plat R1535678)
Unincorporated Arapahoe County · zoning R-P PUD (Z75-030 / P75-027 / P75-028)
Lot 6,401 SF (GIS) · corner lot · flood zone X
House 1977, 2 storey, 1,704 SF + 660 SF basement + 441 SF garage (assessor)
HOA: Cherry Creek Vista II

LOT COVERAGE

Lot area	6,401 SF
Lot coverage limit	none in the PUD plan
Existing deck (assessor)	48 SF (8' x 6') — removed
New deck	8' x 12' · ~92 SF of frame, 10' above grade

SETBACKS

Elevated decks get no setback projection allowance (Arapahoe LDC) — the deck must meet the setbacks
GIS puts the new deck front ~13' off the west line: passes the 10' standard, not the 20' — confirm the governing column and the rear line (corner lot) with Zoning before submittal
E Berry Ave street side 10' required — ~23'-8" clear; south side 5' — ~33' clear
Measured (red): to the nearest lot line from the new work — GIS, not survey

SITE WORK

· Remove the existing 8' x 6' deck and its two posts / pads; haul off
· Two Diamond Pier DP-50 foundations (no excavation spoils), 6x6 DF #1 posts with 4x4 knee braces
· New ledger on the 2' cantilevered second floor per the engineer's detail; membrane + Z-flash, (4) DTT1Z lateral ties
· No stairs — the deck is reached from the house door only
· Call 811 before the piers; protect the lawn and the patio slab at the north wall

Arapahoe County GIS (address point, parcel, 2024 roof prints, R-P zoning, jurisdiction, flood, wildfire layers), county assessor record and sketch, 2nd Amended PDP P75-027 setback table, county deck permit guide (rev. 8/20/2024) and the county-amended ICC deck guide. GIS lines are not a survey — an ILC governs if Zoning asks.

Plan · approx. 1" = 20' at full size · GIS lines, not a survey

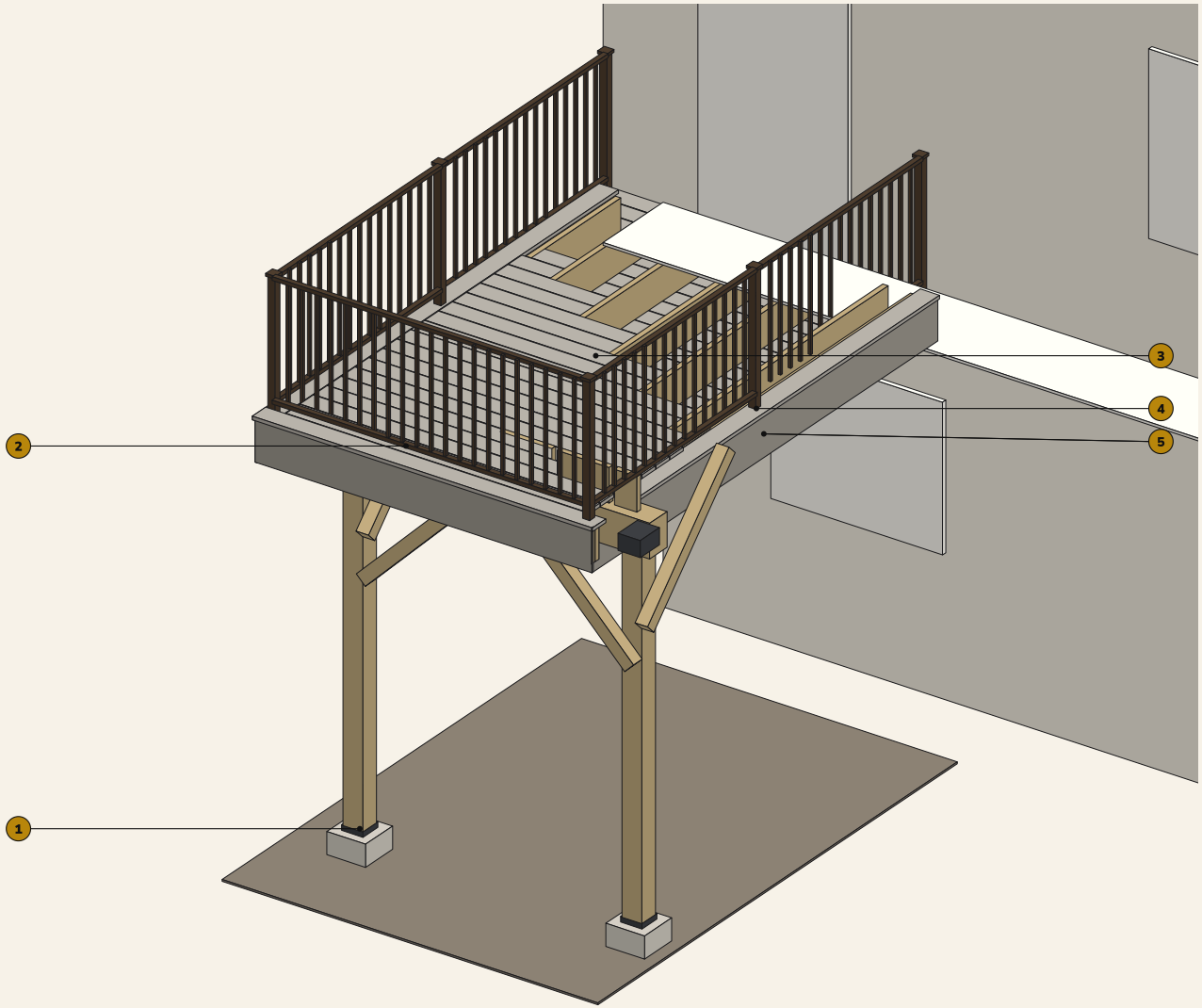
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

A-002 ISOMETRIC — ASSEMBLED, LABELED

Everything you see called out: piers, decking, picture frame, skirt, stair, roof covers, finishes

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 4x4 DF #1 knee brace at 45°, both directions, through-bolted
3. TimberTech Reserve Driftwood field decking, hidden fasteners
4. Picture frame + breaker boards — TimberTech Reserve Driftwood
5. Fascia

House shown for context only (the deck work only). The frame under the decking is called out on S-102 and A-003.

Axonometric · not to scale · the model the plans are drawn from

Elevated deck · A-003 · Isometric — exploded, labeled

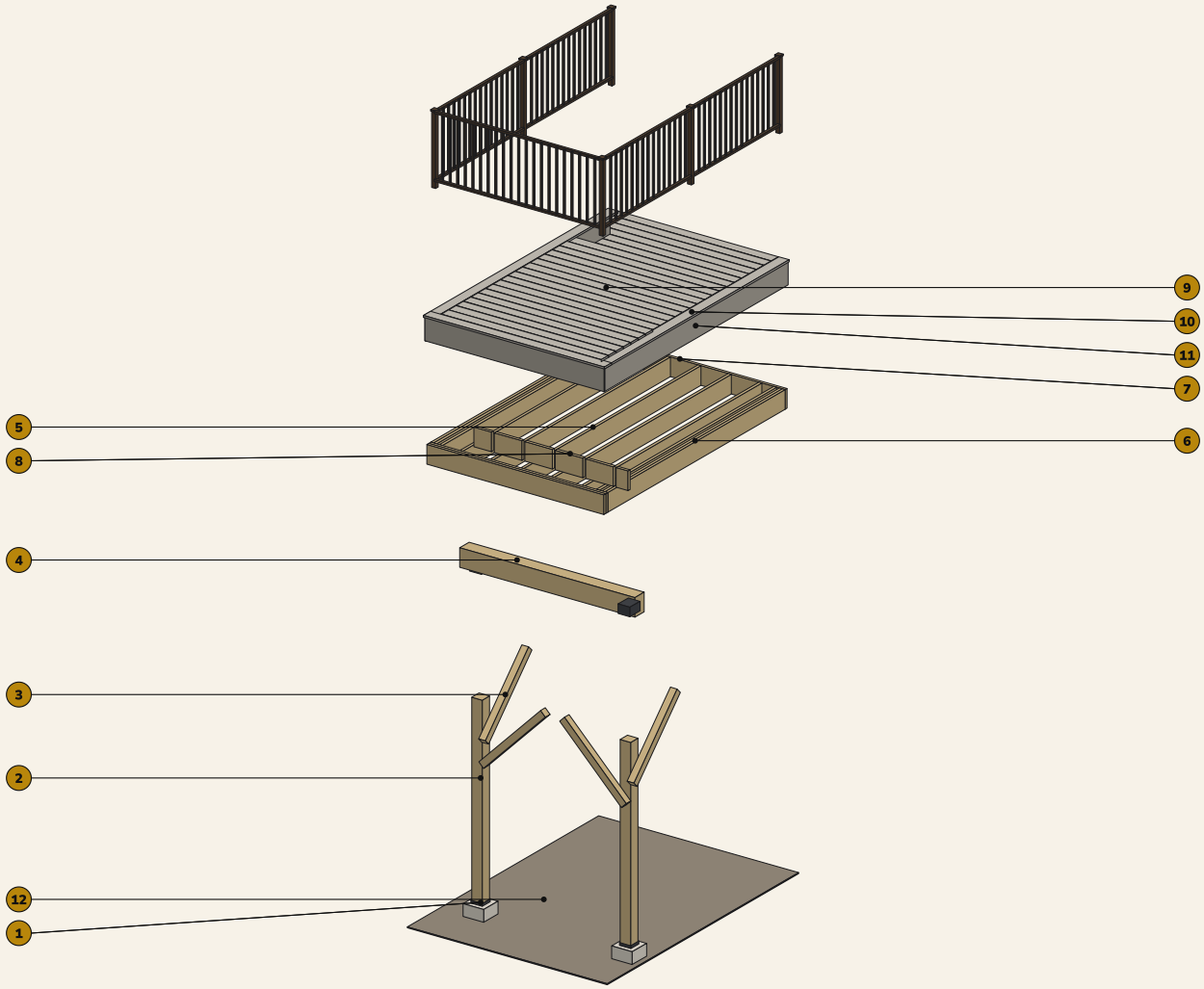
A-003

ISOMETRIC — EXPLODED, LABELED

Each build layer lifted in order: piers · posts + knee braces · beams · frame · decking · rail

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

GSX · Jade Pender 303-550-9558 · October 17, 2026 · Rev 0 · PERMIT SET



KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 6x6 Douglas fir #1 post on a ABA66Z standoff base
3. 4x4 DF #1 knee brace at 45°, both directions, through-bolted
4. Beam
5. 2x10 joist
6. Rim
7. Ledger
8. Blocking
9. TimberTech Reserve Driftwood field decking, hidden fasteners
10. Picture frame + breaker boards — TimberTech Reserve Driftwood
11. Fascia
12. Gravel over 6-mil poly under the decks

Layers lifted in build order. Lifts are for clarity only; every member returns straight down to its place.

Axonometric · not to scale · the model the plans are drawn from

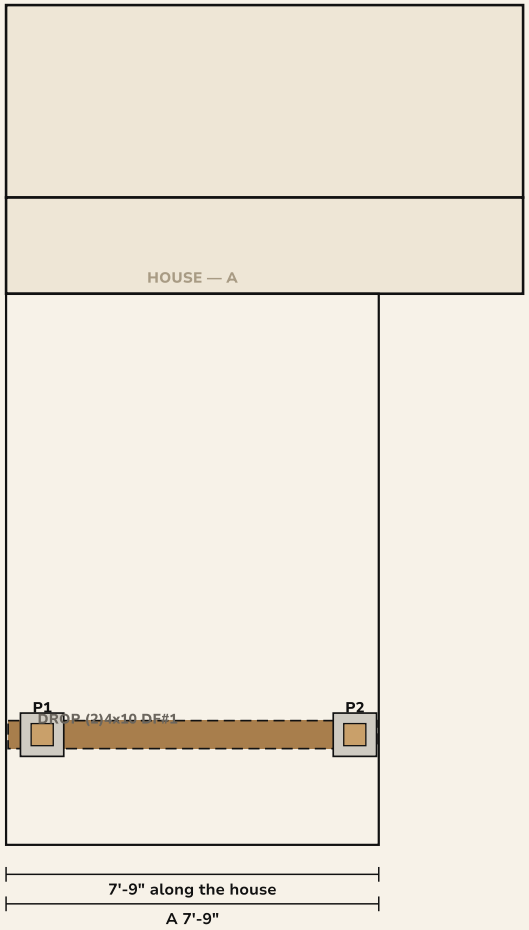
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

S-100 FOUNDATION PLAN

Footings, posts and beam lines with the footing schedule; beams dashed (below the joists)

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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FOOTING SCHEDULE

Diamond Pier DP-50/50 (head + 4 x 50" pins), 3,300 lb allowable

ID	X	Y	ZONE	LOAD
P1	9"	9'-2"	A	2,214 lb
P2	7'-3"	9'-2"	A	2,214 lb

x from the left frame face facing the house; y from the reference wall.
Frost 3'-0" · soil 1,500 psf · worst post 2,214 lb

Plan · not to scale · house at the top · dimensions govern

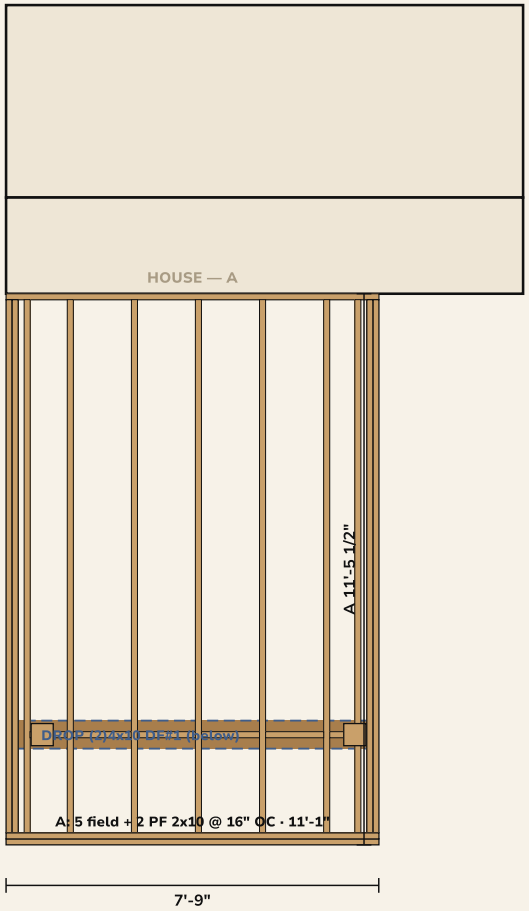
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

S-101 FRAMING PLAN

Ledger, joists, rims, blocking, beams; every member in the model — the cut list is on T-600

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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FRAMING NOTES

7 2x10 #1 True Frame Joist SYP GC @ 16" OC x 11'-1"

2-ply 2x10 front and sides

2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners

blocking one row over the beam, staggered

G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

DROP (2)4x10 DF#1: 7'-8", 2 posts at 9" / 7'-3" (worst 2,214 lb)

Connectors:

1/2" x 10" HDG hex bolt + nut + 2 washers — knee braces (brace to the 6x6 post): 4

1/2" x 12" HDG hex bolt + nut + 2 washers — knee braces (brace to the (2)4x beam): 2

1/2" x 6" HDG hex bolt + nut + 2 washers — knee braces (brace to the joist block): 2

Simpson LUS28Z face-mount hanger 2x8/2x10: 14

Simpson HUCQ210-2-SDS concealed-flange double 2x10 hanger (SDS screws included): 4

Simpson DTT1Z lateral tie kit w/ SDS screws: 4

Simpson H2.5AZ hurricane tie (joists to the drop beam, hidden): 9

Simpson CCQ column cap, 7" seat for the (2)4x10 beam on a 6x6 post (SDS screws incl.) — confirm SKU / engineer's equal:

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

Elevated deck · S-102 · Structure isometric, labeled

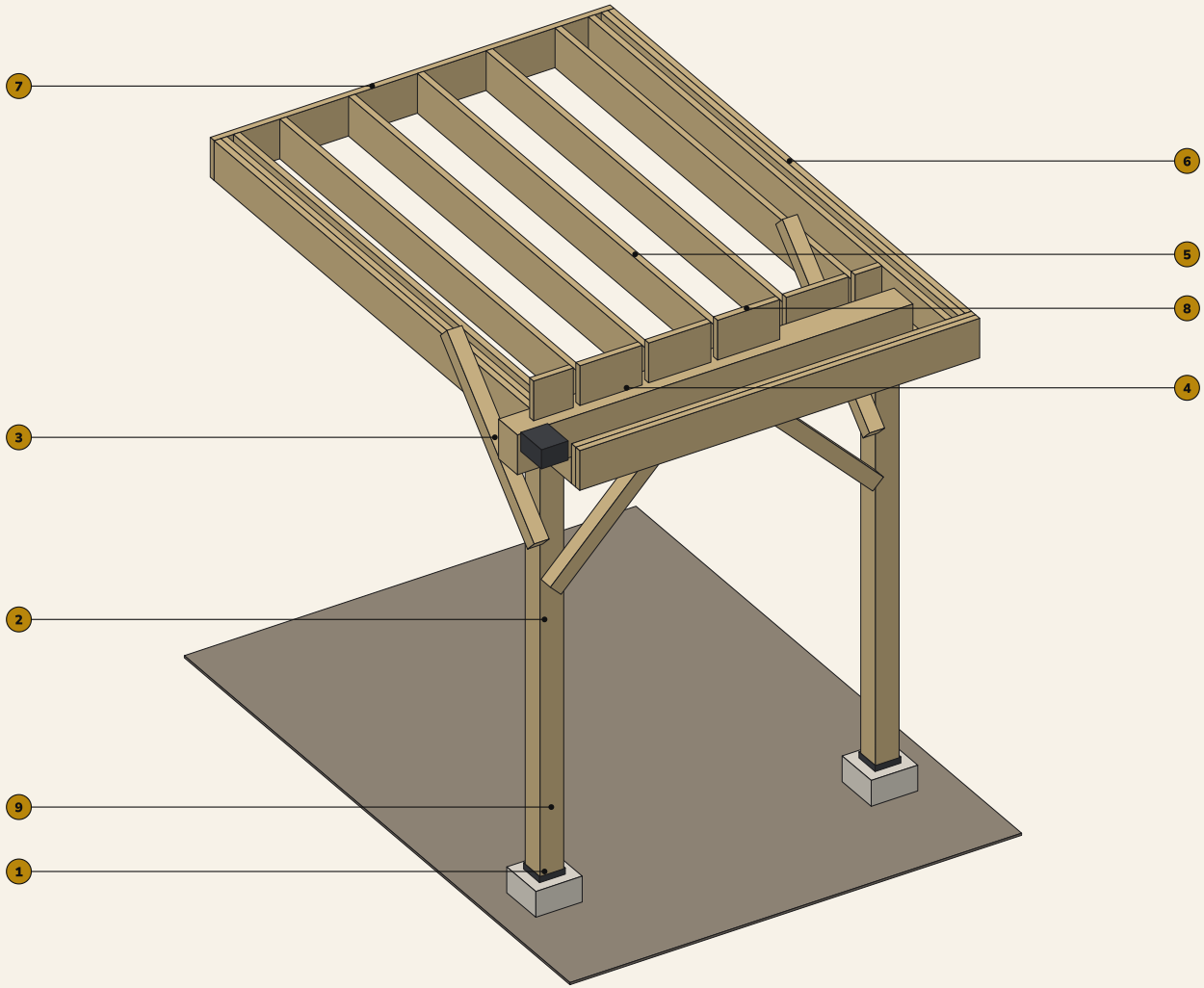
S-102

STRUCTURE ISOMETRIC — FRAME, LABELED

Piers, post bases, posts, knee braces, post caps, beam, joists, double rims, ledger, blocking

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 6x6 Douglas fir #1 post on a ABA66Z standoff base
3. 4x4 DF #1 knee brace at 45°, both directions, through-bolted
4. Beam
5. 2x10 joist
6. Rim
7. Ledger
8. Blocking
9. Gravel over 6-mil poly under the decks

Decking, rail and house removed to show the frame. Joists hang from the ledger in face-mount hangers, bear on the drop beam with H2.5AZ ties; double rims in concealed-flange hangers.

Axonometric · not to scale · the model the plans are drawn from

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

Elevated deck · A-101 · Decking plan

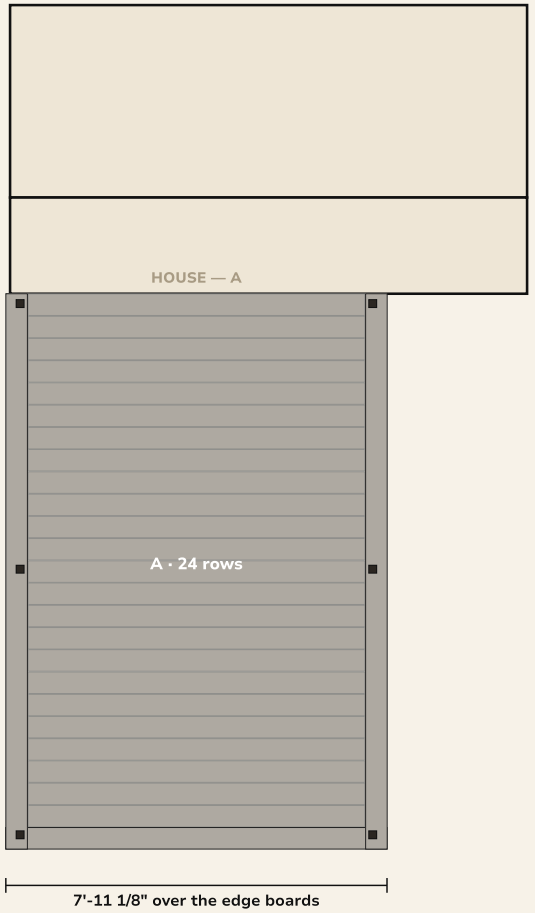
A-101

DECKING PLAN

Every board, border and divider as it lays; rail posts shown for the pockets

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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DECKING

Field: TimberTech Reserve Driftwood, 5.36" x 0.94", 3/16" gaps

Border: Reserve Driftwood

Boards run parallel to the house; edge overhang 1 1/16" over the fascia

CONCEALoc at every joist in every gap (168)

Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then

Rows per zone: A 24 (no rips)

Plan · not to scale · house at the top · dimensions govern

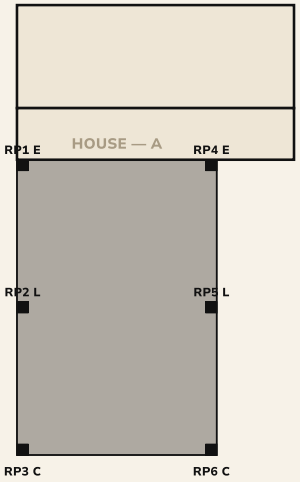
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

A-201 RAILING PLAN + ELEVATION

Post IDs and types (E end · L line · C corner), bay lengths, outer-edge elevation

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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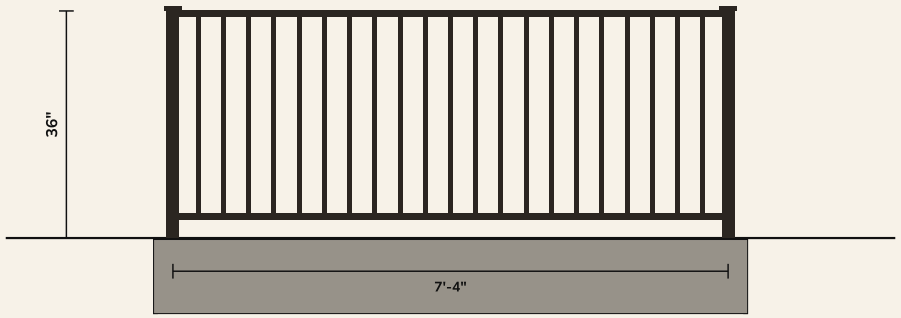
RAILING

IRX Baluster 36" Bronze: 5 bays, 6 posts, 31 LF

IRX Baluster 36" — 6 posts (2 CORNER, 2 END, 2 LINE) surface-mounted on the decking over a 2x blocking pair between the rim plies and the PF joist at every post, (4) HeadLOK 6" per post through the base

Bays: left 5'-6 1/4" · left 5'-6 1/4" · right 5'-6 1/4" · right 5'-6 1/4" · front 7'-4"

Guard 36" min above the deck; openings under 4" (IRC R312). Rail LF for labor = deck-edge LF of actual rail.



Plan above · elevation of the outer edge below

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

Elevated deck · A-202 · Elevations

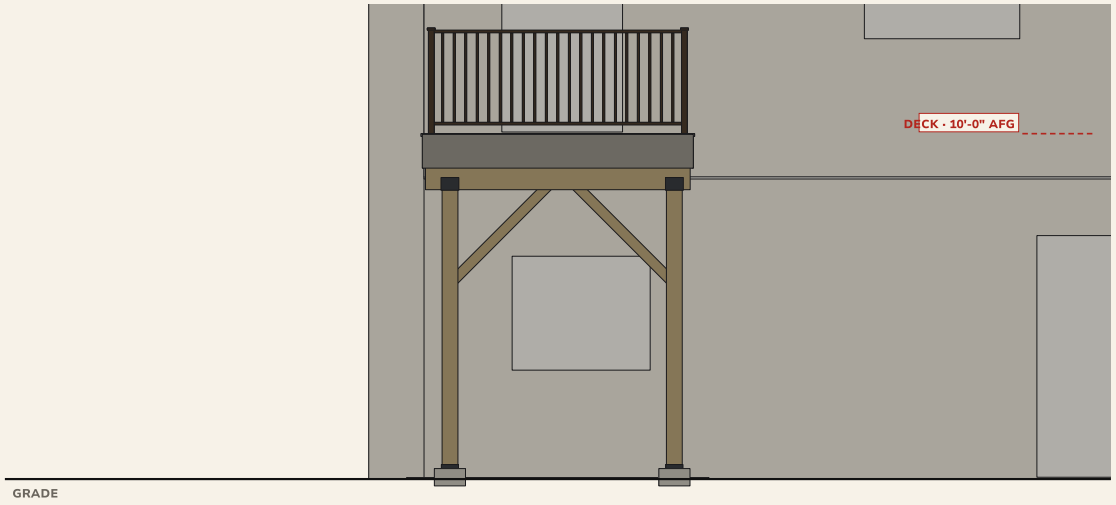
A-202 **ELEVATIONS**

Front from the yard and both ends, drawn from the model — deck and roof-cover heights

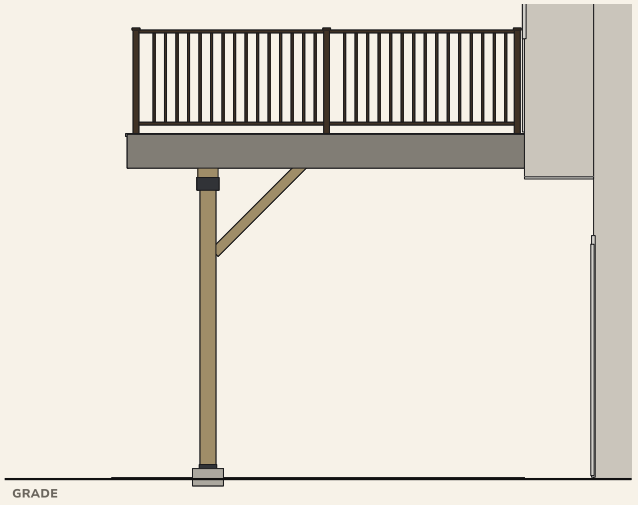
5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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FRONT ELEVATION — from the yard



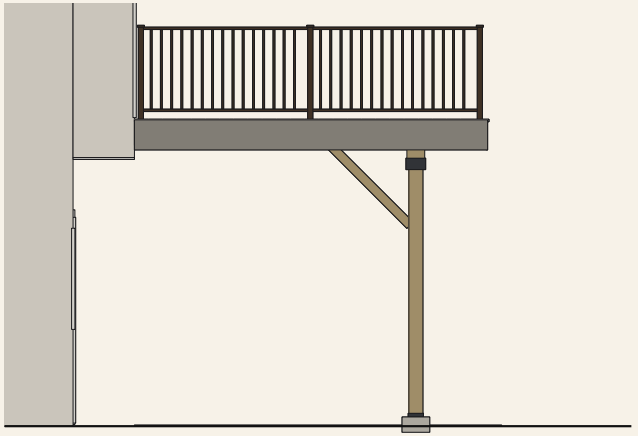
RIGHT END ELEVATION



NOTES

· Deck surface 10'-0" above grade
· Heights AFG = above finished grade at the beam line; field-verify grade before setting the piers
· Deck frame: 2x10 #1 True Frame GC @ 16" OC on a (2)4x10 DF #1 drop beam, 2 posts with 4x4 knee braces both ways
· Guard 36" min. on every open side — the surface is 10'-0" above grade (IRC R312.1)

LEFT END ELEVATION



Elevations · orthographic · not to scale · heights govern

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

Elevated deck · A-301 · Section + front elevation

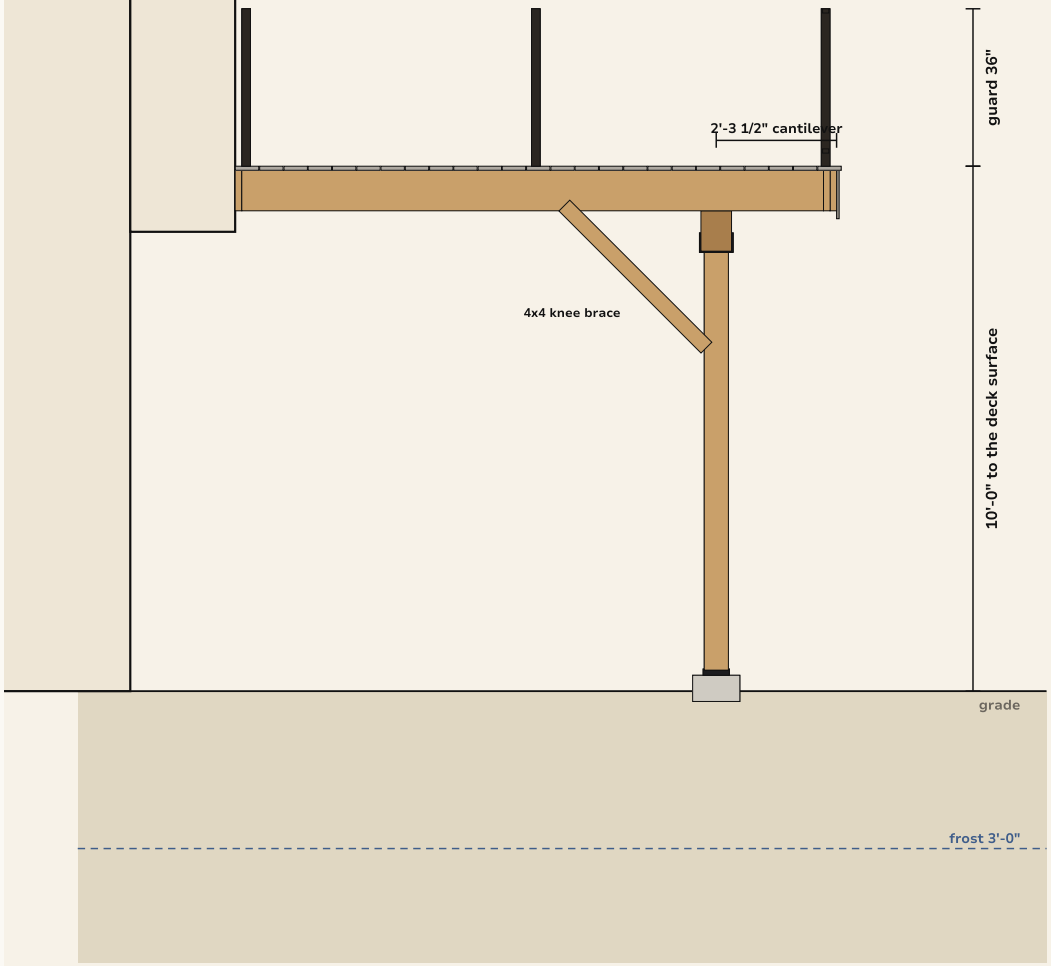
A-301 — **SECTION + FRONT ELEVATION**

Cut through the deepest zone; heights to grade, frost line, cantilever, guard

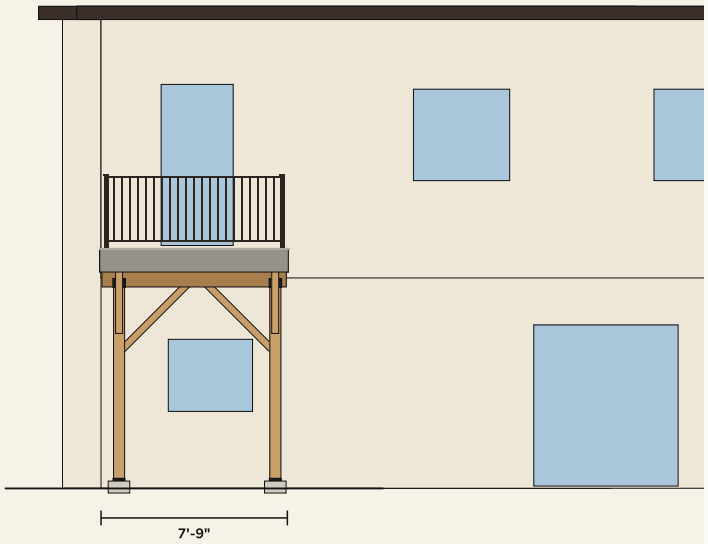
5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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SECTION through zone A at 3'-10 1/2" — looking toward the left end



FRONT ELEVATION — from the yard



SECTION NOTES

Deck surface 10'-0" above grade; guard 36" where the drop exceeds 30"

DROP (2)4x10 DF#1: 7'-8", 2 posts at 9" / 7'-3" (worst 2,214 lb)

Posts 2 x 6x6 Douglas fir #1 with 4x4 knee braces on DP-50/50

2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners

G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

Section left · elevation right · not to a common scale

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

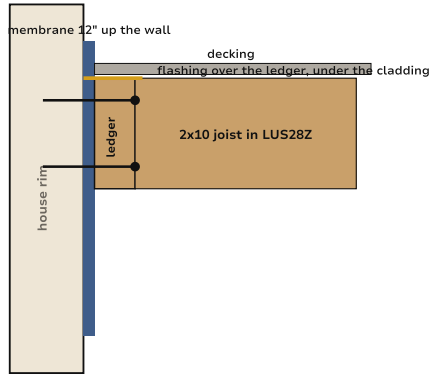
D-501 DETAILS

Ledger · post / beam / cap · footing · rail post · deck edge · the job's special condition

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

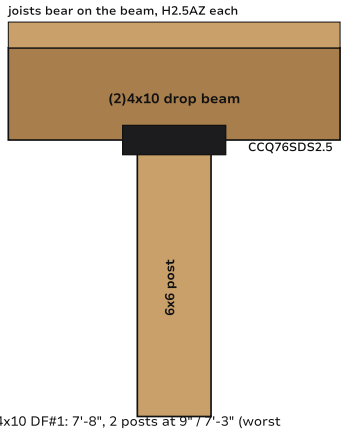
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1 LEDGER AT THE HOUSE



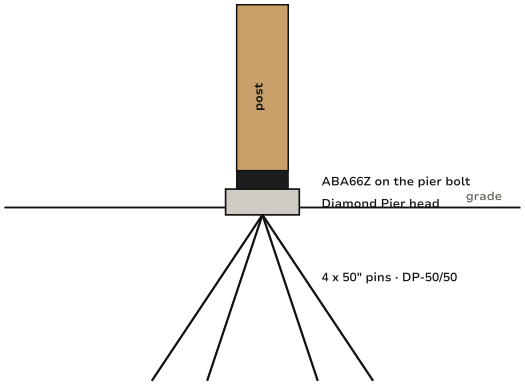
2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners

2 POST · BEAM · CAP

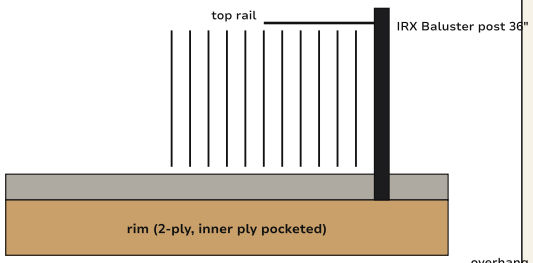


DROP (2)4x10 DF#1: 7'-8", 2 posts at 9" / 7'-3" (worst 2,214 lb)

3 FOOTING · POST BASE

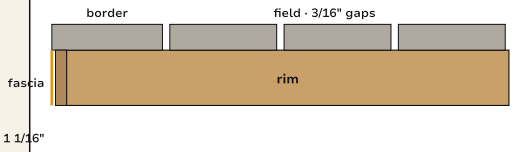


4 RAIL POST · GUARD



IRX Baluster 36" — 6 posts (2 CORNER, 2 END, 2 LINE) surface-mounted on the decking over a 2x blocking pair between the rim plies and the PF joist at every post, (4) HeadLOK 6" per post through the base

5 DECK EDGE · BORDER



CONCEALoc at every joist in every gap (168)

6 BLOCKING

Blocking one row over the beam, staggered up and down so every block can be end-nailed.

Elevated deck · D-502 · Connection details — frame

D-502

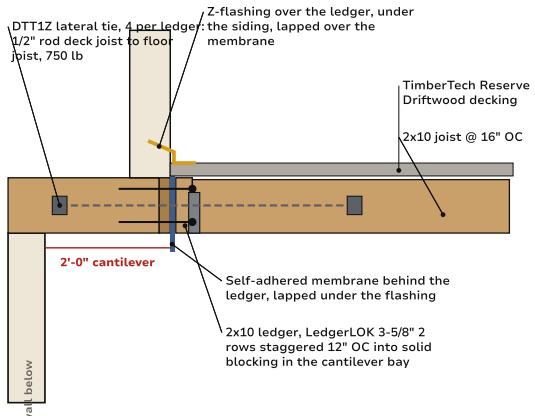
CONNECTION DETAILS — FRAME

Ledger on the cantilever · hangers · joists on the beam · (2)4x10 on the cap · pier + base · knee braces

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

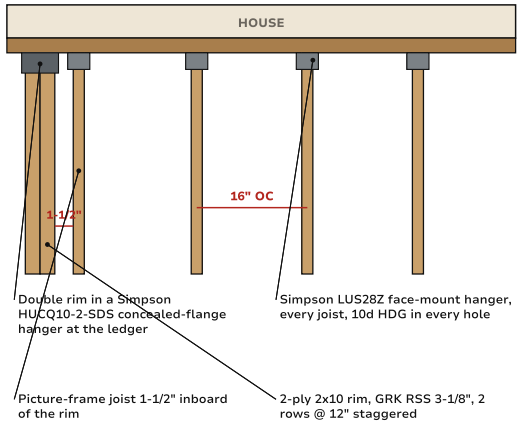
GSX · Jade Pender 303-550-9558 · October 17, 2026 · Rev 0 · PERMIT SET

1 LEDGER ON THE CANTILEVERED FLOOR — SECTION

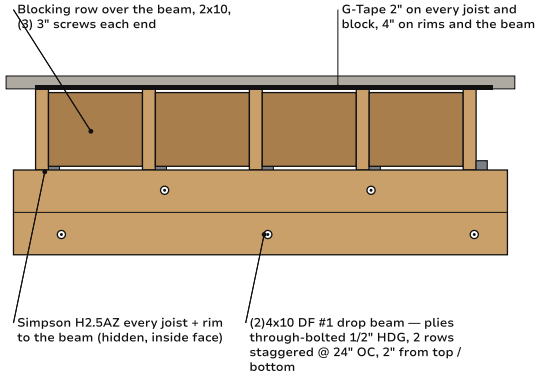


ENGINEER: Arapahoe County bars a ledger on a cantilevered floor without engineer approval. The stamped detail governs: ledger into the blocked cantilever joists, or a freestanding beam on posts at the house.

2 HANGERS — PLAN AT THE LEDGER + SIDE RIM



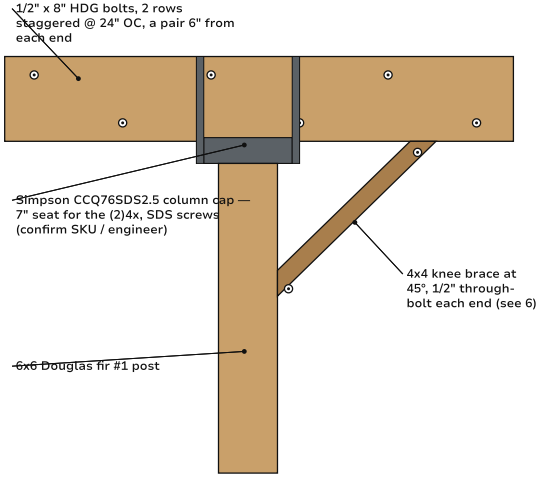
3 JOISTS ON THE DROP BEAM — SECTION



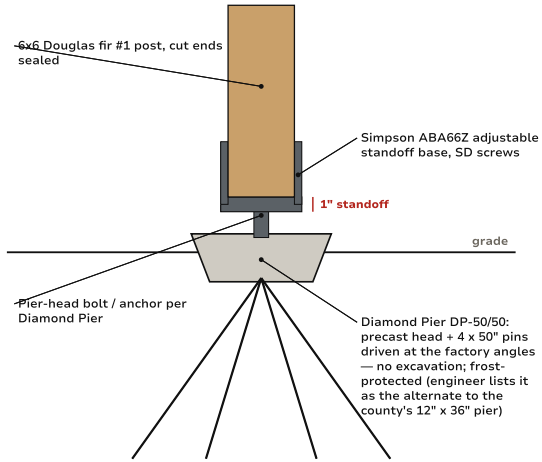
Joists bear the full beam width

Cantilever 2'-3 1/2" past the beam — over L/4: engineer

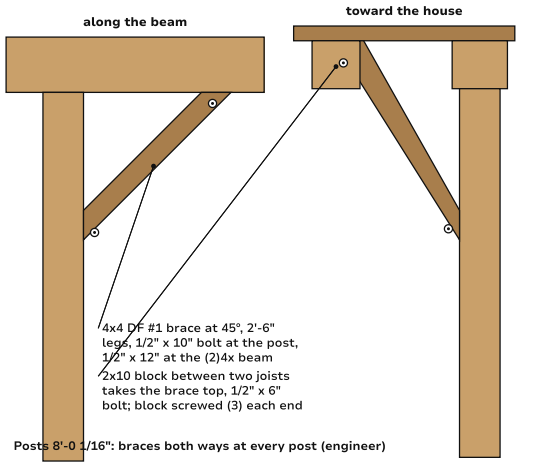
4 (2)4x10 BEAM ON THE 6x6 — ELEVATION



5 POST BASE ON THE DIAMOND PIER — SECTION



6 KNEE BRACES — BOTH DIRECTIONS



Posts 8'-0 1/16": braces both ways at every post (engineer)

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

Elevated deck · D-504 · Connection details — deck edge, rail, lateral

D-504

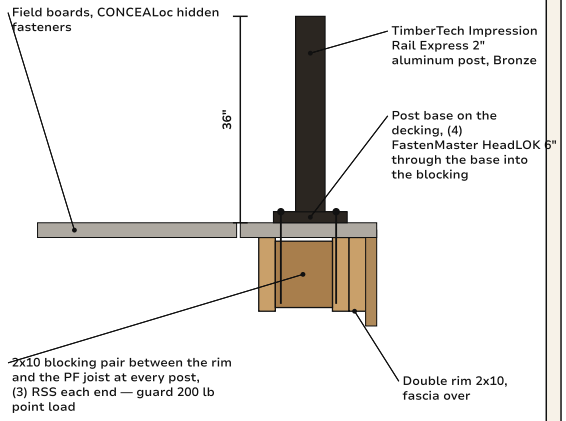
CONNECTION DETAILS — DECK EDGE, RAIL, LATERAL

Rail post mounting · picture frame + fascia · guard bay · DTT1Z lateral ties · joist tape · fastening schedule

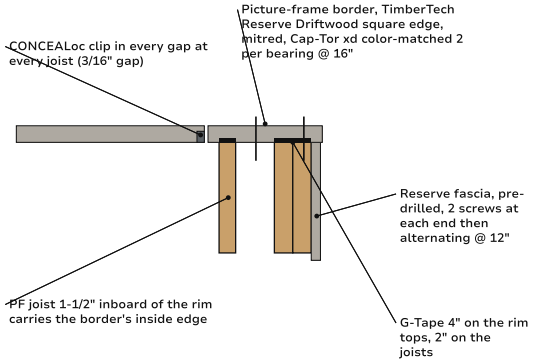
5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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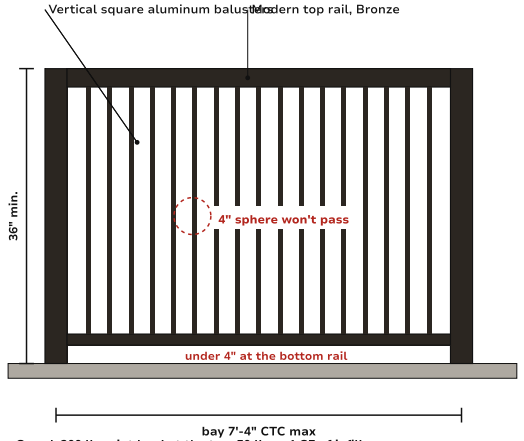
1 RAIL POST — SURFACE MOUNT ON BLOCKING



2 DECK EDGE — PICTURE FRAME + FASCIA

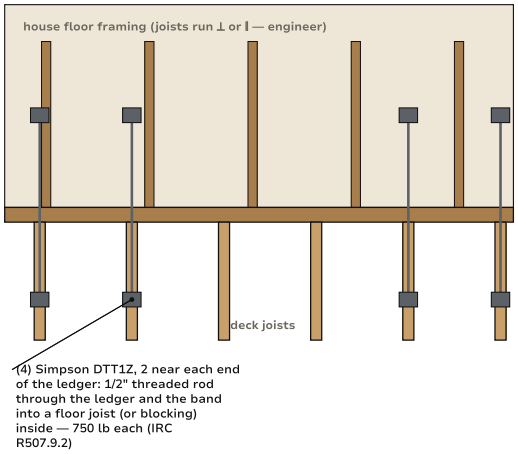


3 GUARD — ONE BAY, ELEVATION

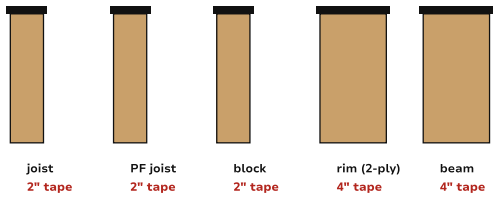


Guard: 200 lb point load at the top, 50 lb on 1 SF of infill (IRC Table R301.5) — submit the rail code report

4 LATERAL LOAD — DTT1Z TIES (PLAN)



5 JOIST TAPE + BLOCKING



Nichigo G-Tape 3035 (black) on every top edge before decking, lapped over the ends and down the sides 1/2"; one row of 2x10 blocking over the beam, (3) structural screws each

6 FASTENING SCHEDULE

Ledger to house	LedgerLOK 3-5/8", 2 rows staggered 12" OC — engineer
Lateral	(4) DTT1Z, 1/2" rod into the floor framing
Joist to ledger	LUS28Z, 10d HDG every hole
Rim to ledger	HUCQ10-2-SDS, SDS screws
Rim plies	GRK RSS 3-1/8", 2 rows @ 12" staggered
Joist to beam	H2.5AZ each joist + rim, 10d x 1-1/2"
Beam plies	1/2" x 8" HDG bolts, 2 rows staggered @ 24"
Beam to post	CCQ76SDS2.5 cap, SDS screws
Post to pier	ABA66Z on the Diamond Pier, SD screws
Knee braces	1/2" through-bolt each end (x10 / x12 / x6)
Field boards	CONCEALoc, every joist, every gap
Borders / fascia	Cap-Tor xd color-matched, pre-drilled
Rail posts	(4) HeadLOK 6" into blocking

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

Deck



From the yard



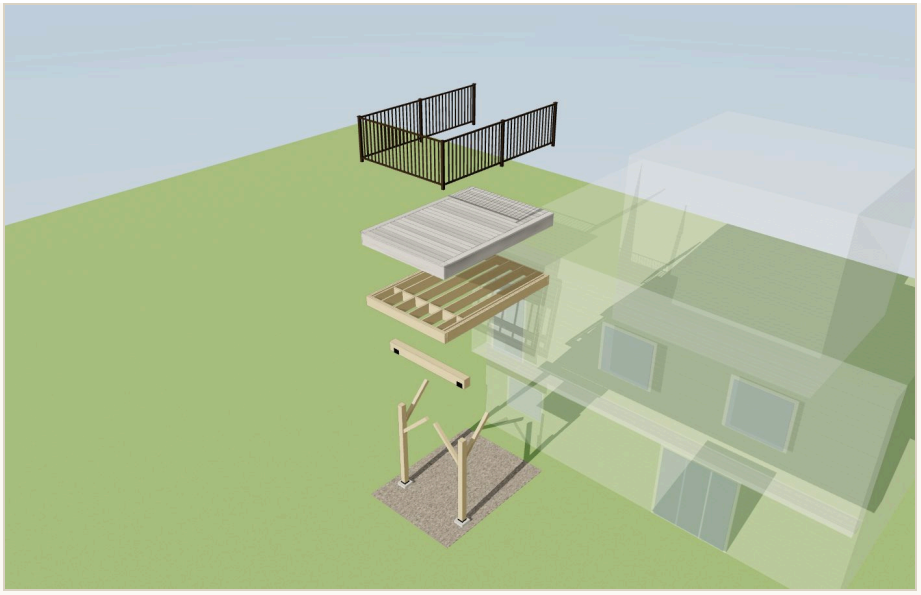
From the front corner



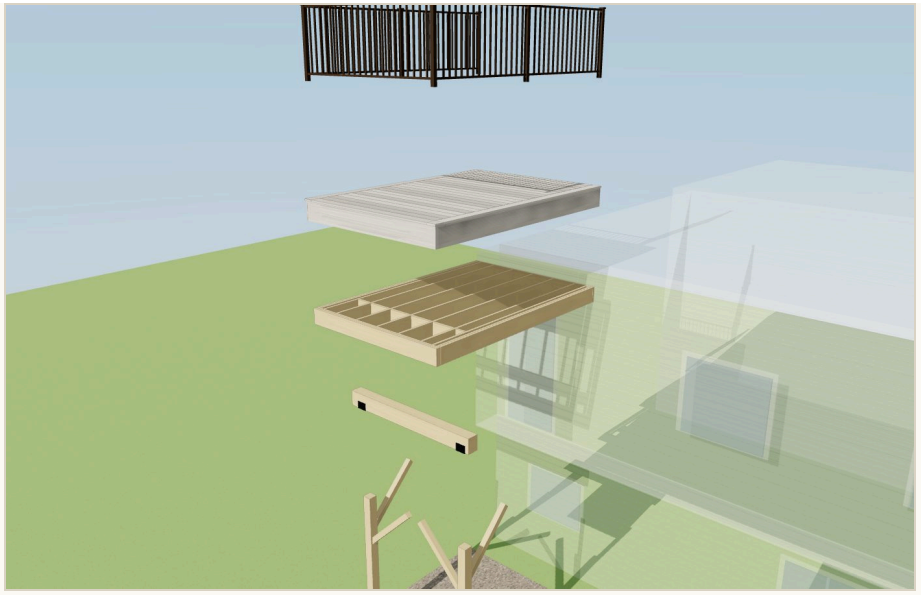
Isometric



Isometric — opposite corner



Exploded — every layer



Exploded — from the corner



Structure from below



Under the deck